1014 U2 intro to probability

1. Each accountant at a large accounting firm was classified according to accountant level (junior or senior) and method of transportation to work (walk, bus, drive alone, or carpool). The responses of the 320 accountants at the firm are summarized in the table.

	Junior	**Senior**	**Total**
Walk	25	3	28
Bus	87	12	99
Drive alone	96	25	121
Carpool	52	20	72
Total	260	60	320

What proportion of the accountants at the firm are at the senior level and carpool to work?

(A) $\frac{20}{60}$

(B) $\frac{20}{72}$

(C) $\frac{20}{320}$ ✓

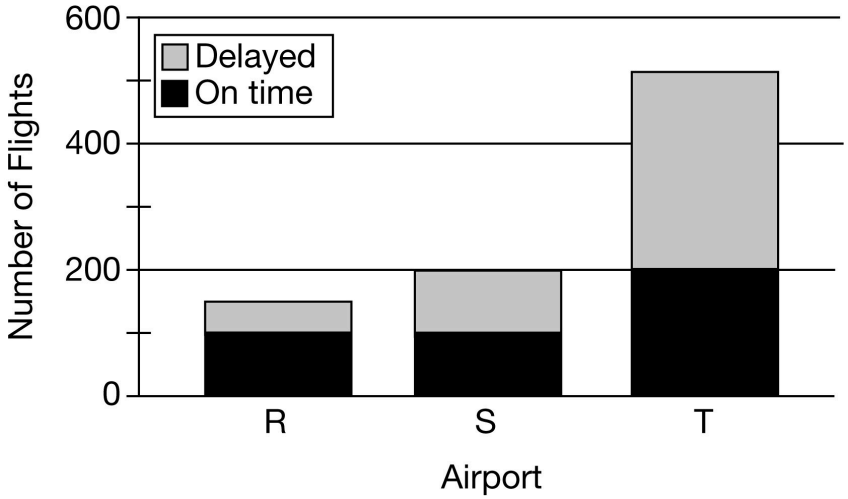
(D) $\frac{112}{320}$

Answer C

Correct. The proportion of accountants who are both senior level and carpool to work is $\frac{20}{320}$.

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2. The following segmented bar chart shows the number of flights that were either on time or delayed at three different airports on one day.



Which of the following statements is supported by the bar chart?

(A) Airport T has the greatest percentage of on-time flights compared to the other two airports.

(B) The number of on-time flights at Airport S is half the number of on-time flights at Airport T. ✓

(C) The number of on-time flights at Airport R is less than the number of on-time flights at Airport S.

(D) The number of flights at Airport T is equal to the total number of flights at Airports R and S combined.

Answer B

Correct. Airport S has 100 on-time flights, and Airport T has 200 on-time flights. Since 100 is one-half of 200, Airport S has one-half the number of on-time flights that Airport T has.

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3.

	Job	No Job	Total
Juniors	13	5	18
Seniors	13	26	39
Total	26	31	57

A survey of 57 students was conducted to determine whether or not they held jobs outside of school. The two-way table above shows the number of students by employment status (job, no job), and class (juniors, seniors). Which of the following best describes the relationship between employment status and class?

(A) There appears to be no association, since the same number of juniors and seniors have jobs.

(B) There appears to be an association, since there are more seniors than juniors in the survey.

(C) There appears to be an association, since the proportion of juniors having jobs is much larger than the proportion of seniors having jobs. ✓

(D) A measure of association cannot be determined from these data.

4. A financial analyst reports that for people who work in the finance industry, the probability that a randomly selected person will have a tattoo is 0.20.

Which of the following is the best interpretation of the probability 0.20 ?

(A) For all workers in the United States, 20% will work in finance.

(B) For all finance workers, 20% will have a tattoo. ✓

(C) For all people with tattoos, 20% will work in finance.

(D) For a specific group of 5 finance workers, 1 will have a tattoo.

Answer B

Correct. In the long run, 0.20 is the relative frequency at which a finance worker will have a tattoo.

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5. A company blood drive collected 42 pints of donated blood. The table shows the classification of each of the pints based on blood type (O, A, B, or AB) and Rh factor (positive or negative).

Blood Type Classification

	Type O	Type A	Type B	Type AB	Total
Negative	3	4	1	0	8
Positive	16	13	4	1	34
Total	19	17	5	1	42

Based on the table, of all pints of blood classified as positive, approximately what proportion is blood type A?

(A) 0.31

(B) 0.38 ✓

(C) 0.50

(D) 0.76

Answer B

Correct. The correct proportion uses the cell for Type A in the positive row divided by the total of all blood types in the positive row $\frac{13}{(16+13+4+1)} = 0.38$.

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6. At a sporting event, cheerleaders will throw 50 bundled T-shirts into the crowd. The T-shirt sizes consist of 10 small, 15 medium, and the remainder either large or extra large. Suppose Ana catches a T-shirt. What is the probability that she will catch a T-shirt that is not a size small?

(A) 0.20

(B) 0.50

(C) 0.67

(D) 0.80 ✓

Answer D

Correct. There are 50 total T-shirts and 10 are small, so 40 are not small. The probability is thus $\frac{40}{50} = 0.80$.

7. A middle school chess club has 5 members: Adam, Bradley, Carol, Dave, and Ella. Two students from the club will be selected at random to participate in the county chess tournament. What is the probability that Adam and Ella will be selected?

(A) $\frac{1}{20}$

(B) $\frac{1}{10}$ ✓

(C) $\frac{1}{8}$

(D) $\frac{1}{4}$

Answer B

Correct. The two players selected could be AB, AC, AD, AE, BC, BD, BE, CD, CE, or DE, so there are 10 equally likely possible outcomes. Alternatively, there are $\binom{5}{2} = 10$ outcomes when 2 people are chosen from 5 people. One of these outcomes is AE, so the probability is $\frac{1}{10}$.

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8. A prize game at a circus has a spinner with 12 sections, where each section is equally likely to be selected when the spinner is spun. There are 7 blue sections and 5 orange sections. Camille has spun the spinner 4 times and the pattern of outcomes was orange, blue, orange, blue. Camille will spin the spinner two more times.

Which of the following statements is true?

(A) On Camille's fifth spin of the spinner, the probability that the spinner will land in a blue-colored section is equal to the probability that the spinner will land in an orange-colored section.

(B) On Camille's fifth spin of the spinner, the probability that the spinner will land in a blue-colored section is less than the probability that the spinner will land in an orange-colored section.

(C) On Camille's fifth and sixth spins of the spinner, the probability that the spinner will land in a blue-colored section for both spins is equal to the probability that the spinner will land in an orange-colored section for both spins.

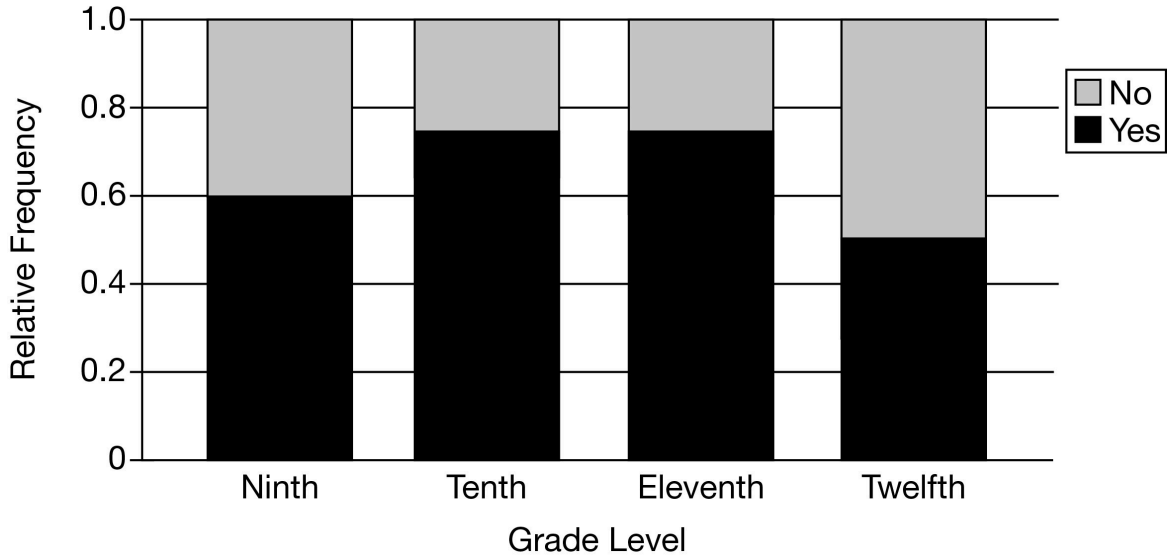
(D) On Camille's fifth and sixth spins of the spinner, the probability that the spinner will land in a blue-colored section for both spins is greater than the probability that the spinner will land in an orange-colored section for both spins. ✓

Answer D

Correct. Because spins are independent, the probability that the spinner will land in a blue-colored section on both the fifth and sixth spins, $\left(\frac{7}{12}\right)\left(\frac{7}{12}\right) = \frac{49}{144}$, is greater than the probability that the spinner will land in an orange-colored section on both the fifth and sixth spins, $\left(\frac{5}{12}\right)\left(\frac{5}{12}\right) = \frac{25}{144}$.

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9. The following bar chart displays the relative frequency of responses of students, by grade level, when asked, "Do you volunteer in a community-service activity?"



Which of the following statements is not supported by the bar chart?

(A) More than 60% of both tenth-grade and eleventh-grade students responded yes.

(B) Less than 40% of tenth-grade students responded no.

(C) The number of tenth-grade students who responded yes was greater than the number of ninth-grade students who responded yes. ✓

(D) The percentage of eleventh-grade students who responded no was less than the percentage of ninth-grade students who responded no.

Answer C

Correct. The statement is not supported by the bar chart. The graph only gives information about the *percentage* of students who responded yes or no. The graph gives no information on the *number* of students who responded.

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10. The director of a technical school was curious about whether there is a relationship between students who complete one of the school's most popular health sciences certificate programs and whether those students go on to complete more advanced studies in the health sciences within two years of completing the certificate program. She randomly selected 100 students who completed the program. Data collected on these students are shown in the table below.

		Completed More Advanced Studies		
		Yes	No	Total
Completed Most Popular Health Sciences Certificate Program	Yes	35	25	60
	No	5	35	40
	Total	40	60	100

Which of the following statements is true for these 100 students?

(A) Being a person who completed the program is less likely than being a person who did not complete the program.

(B) Being a person who completed the program and completed more advanced studies is less likely than being a person who did not complete the program and did not complete more advanced studies.

(C) Being a person who did not complete the program but completed more advanced studies is less likely than being a person who completed the program and completed more advanced studies. ✓

(D) Being a person who completed the program but did not complete more advanced studies is more likely than being a person who did not complete the program and did not complete more advanced studies.

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11. A sample of 942 homeowners are classified, in the two-way frequency table below, by the number of credit cards they have and the number of years they have owned their current homes.

	Number of Years Owning Current Home			
Number of Credit Cards	Less than One Year	One Year to Three Years	More than Three Years	Total
One	265	53	16	334
Two	84	19	4	107
Three	201	68	20	289
Four or more	134	58	20	212
Total	684	198	60	942

Of the homeowners in the sample who have four or more credit cards, what proportion have owned their current homes for at least one year?

(A) 78/212 ✓

(B) 78/258

(C) 78/942

(D) 258/942

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12. At a local ice-cream store, 210 people were surveyed on whether they preferred eating ice cream from a cone or a cup. Of the 210 people surveyed, 70 were adults and 140 were children. Of the responses, 150 indicated the cone as the preferred method of eating ice cream. For those surveyed, there was no association between age and preferred method of eating ice cream.

Which of the following tables shows the distribution of responses?

(A)

	Cone	Cup	Total
Adults	35	35	70
Children	115	25	140
Total	150	60	210

(B)

	Cone	Cup	Total
Adults	40	30	70
Children	110	30	140
Total	150	60	210

(C) ✓

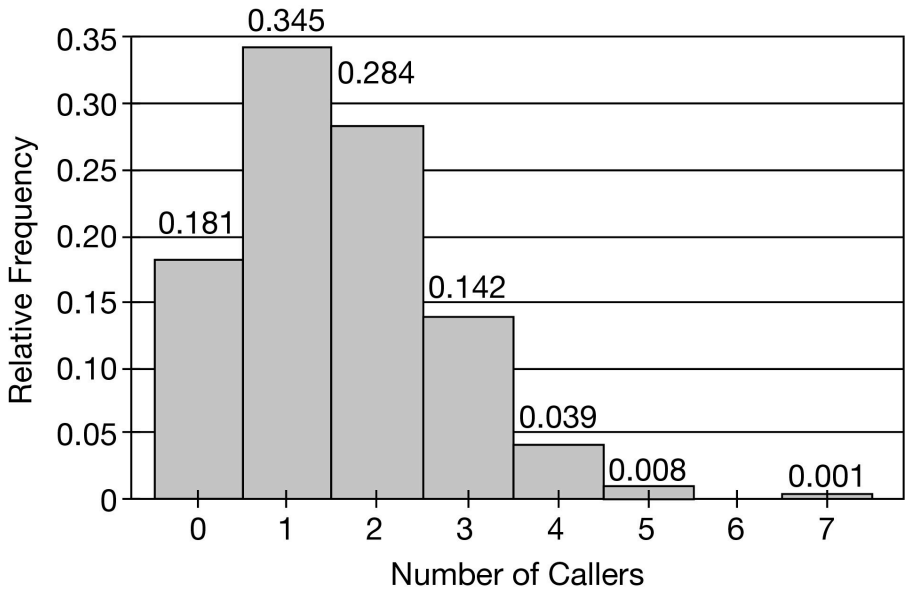
	Cone	Cup	Total
Adults	50	20	70
Children	100	40	140
Total	150	60	210

(D)

	Cone	Cup	Total
Adults	65	5	70
Children	85	55	140
Total	150	60	210

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13. An online customer service department estimates that about 15 percent of callers have to wait more than 8 minutes to have their calls answered by a person. The department conducted a simulation of 1,000 trials to estimate the probabilities that a certain number of callers out of the next 10 callers will have to wait more than 8 minutes to have their calls answered. The simulation is shown in the following histogram.



Based on the simulation, what is the probability that at most 2 of the next 10 callers will have to wait more than 8 minutes to have their calls answered?

(A) 0.190

(B) 0.474

(C) 0.526

(D) 0.810 ✓

Answer D

Correct. The phrase "at most 2" means 0, 1, or 2. The sum of the relative frequencies for the bars corresponding to 0, 1, and 2 callers is $0.181 + 0.345 + 0.284 = 0.810$.

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14. A local company is interested in supporting environmentally friendly initiatives such as carpooling among employees. The company surveyed all of the 200 employees at the downtown offices. Employees responded as to whether or not they own a car and to the location of the home where they live. The results are shown in the table below.

		Location of Home			
		Downtown Area In the City	Elsewhere In the City	Outside the City	Total
Car Ownership	Yes	10	15	35	60
	No	60	55	25	140
	Total	70	70	60	200

Which of the following statements about a randomly chosen person from these 200 employees is true?

(A) If the person owns a car, he or she is more likely to live elsewhere in the city than to live in the downtown area in the city. ✓

(B) If the person does not own a car, he or she is more likely to live outside the city than to live in the city (downtown area or elsewhere).

(C) The person is more likely to own a car if he or she lives in the city (downtown area or elsewhere) than if he or she lives outside the city.

(D) The person is more likely to live in the downtown area in the city than elsewhere in the city.

15. A fair die with its faces numbered from 1 to 6 will be rolled. Which of the following is the best interpretation of the probability that the number landing face up will be less than 3 ?

(A) For many rolls of the die, the long-run relative frequency of a number less than 3 landing face up is $\frac{1}{3}$. ✓

(B) For many rolls of the die, the long-run relative frequency of a number less than 3 landing face up is $\frac{1}{2}$.

(C) For many rolls of the die, the long-run relative frequency of a number less than 3 landing face up is $\frac{2}{3}$.

(D) For three rolls of the die, a number less than 3 will land face up one time.

Answer A

Correct. A roll of a fair die can result in 6 possible outcomes. There are 2 outcomes with a number less than 3 (1 and 2). In the long run, the relative frequency of rolling a number less than 3 is $\frac{2}{6} = \frac{1}{3}$.

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16. The following data were collected from a random sample of people, who identified their favorite type of juice. The results are shown in the following two-way table.

	Orange	Grape	Cranberry	Apple	Total
Adults	600	500	50	150	1,300
Children	400	100	50	150	700
Total	1,000	600	100	300	2,000

What proportion of the children identified orange as their favorite type of juice?

(A) $\frac{400}{1,000}$

(B) $\frac{400}{700}$ ✓

(C) $\frac{400}{2,000}$

(D) $\frac{1,000}{2,000}$

Answer B

Correct. There are 700 children, and 400 of the 700 children identified orange as their favorite type of juice, so $\frac{400}{700}$ is the proportion of children who identified orange as their favorite type of juice.

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17. The following data were collected from a random sample of people on their favorite types of leisure activities and their age. The results are shown in the two-way table below.

	Reading	Exercise	Playing Video Games	Watching Television	Total
0 to 6 years	10	50	100	140	300
7 to 12 years	200	300	200	200	900
13 to 18 years	600	500	100	100	1,300
19 years or greater	500	250	50	200	1,000
Total	1,310	1,100	450	640	3,500

What proportion of the people aged 7 to 18 years gave watching television as their favorite type of leisure activity?

(A) $\frac{300}{2,200}$ ✓

(B) $\frac{200}{900}$

(C) $\frac{100}{1,300}$

(D) $\frac{300}{640}$

Answer A

Correct. There are $900 + 1,300 = 2,200$ people aged 7 to 18 years. 200 of those aged 7 to 12 years and 100 of those aged 13 to 18 years gave watching television as their favorite leisure activity, so $200 + 100 = 300$ of the people aged 7 to 18 years gave watching television as their favorite leisure activity. Thus the proportion of people aged 7 to 18 years who gave watching television as their favorite leisure activity is $\frac{300}{2,200}$.

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18. Mateo plays on his school basketball team. From past history, he knows that his probability of making a basket on a free throw is 0.8. Suppose he wants to create a simulation using random numbers to estimate the probability of making at least 3 baskets on his next 5 free throw attempts. Which of the following assignments of the digits 0 to 9 could be used for the simulation?

(A) Let the digits 0 and 1 represent making a basket and the digits from 2 to 9 represent not making a basket.

(B) Let the digits from 0 to 3 represent making a basket and the digits from 4 to 9 represent not making a basket.

(C) Let the digits from 0 to 6 represent making a basket and the digits from 7 to 9 represent not making a basket.

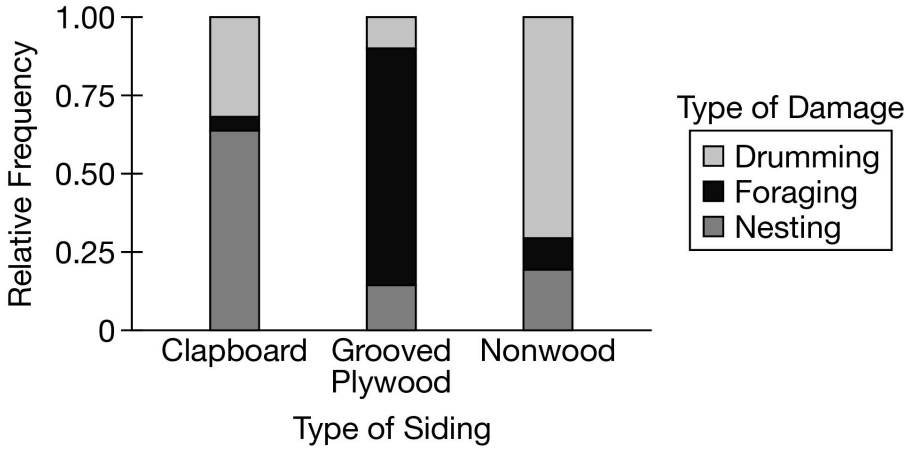
(D) Let the digits from 0 to 7 represent making a basket and the digits 8 and 9 represent not making a basket. ✓

Answer D

Correct. The eight digits 0 to 7 represent 80% of the 10 digits. This matches Mateo's probability of making a basket.

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19. A study was conducted on three types of home siding and the type of damage done to the siding by woodpeckers. Each hole made by a woodpecker was classified as either drumming (territorial signaling), foresting (looking for food), or nesting. The following bar chart shows the relative frequency of the holes for each type of siding.



Which of the following statements is supported by the bar chart?

(A) The proportion of holes created for drumming is greatest for grooved plywood.

(B) The proportion of holes created for drumming is least for grooved plywood. ✓

(C) The number of holes created for drumming is least for grooved plywood.

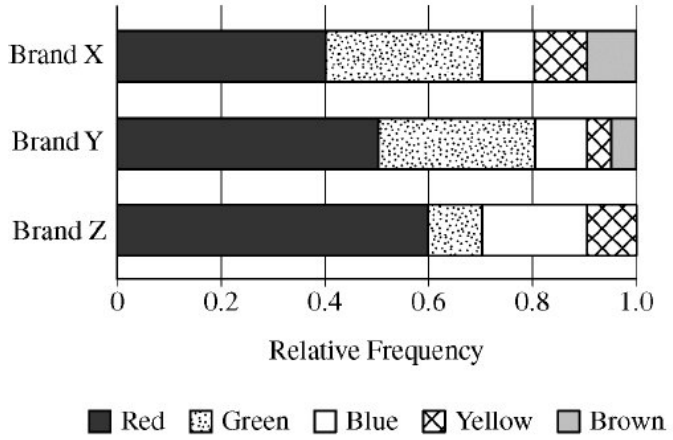
(D) The number of holes created for drumming is greatest for nonwood.

Answer B

Correct. The proportion of holes created by woodpeckers for drumming is around 30% for clapboard, 10% for grooved plywood, and 70% for nonwood. Therefore, the proportion of holes created for drumming is least for grooved plywood.

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20. Three brands of candy pieces—X, Y, and Z—are made in many colors. Shaela bought one bag of each brand and counted the number of pieces of each color. The graph below shows the relative frequency distribution of colors for each bag.



Which of the following statements must be true?

(A) For Brand X, there were more green candy pieces than red candy pieces in the bag.

(B) For Brand Y, there were more red candy pieces than green candy pieces in the bag. ✓

(C) There were more green candy pieces in the Brand X bag than were in the Brand Z bag.

(D) The number of blue candy pieces in the Brand Z bag was equal to the sum of the number of blue candy pieces in the other two bags.